

3 Partial Derivatives

Example 3.6.1 Find the direction of greatest slope of the function $f(x, y) = \frac{x}{x^2+y^2}$ at the point $(1, 2)$ and find the slope in that direction.

3.7 Maximum and Minimum Values

Definition 3.7.1 If f is a function of two variables, x and y , we say that f has a (local) minimum at (a, b) if (a, b) lies in the domain of f , and for points (x, y) in the domain of f 'near' (a, b) we have that

$$f(a, b) \leq f(x, y).$$

Similarly, we say that (a, b) is a (local) maximum if

$$f(a, b) \geq f(x, y)$$

for all (x, y) 'near' (a, b) and in the domain of f .

Theorem 3.7.2 If f is a function of two variables, and has local minimum or maximum at (a, b) , and the partial derivatives of f exist at (a, b) , we have that

$$\frac{\partial f}{\partial x}(a, b) = 0 \text{ and } \frac{\partial f}{\partial y}(a, b) = 0.$$

Example 3.7.3 Find the extreme values of the function

$$f(x, y) = x^2 + y^2 - 2x - 6y + 14.$$

Example 3.7.4 Find the critical points of the function

$$f(x, y) = x^2 - y^2.$$

Theorem 3.7.5 (*Second derivative test*) Suppose that f is a function of two variables, whose second partial derivatives are continuous around the critical point (a, b) . Then we define

$$D(a, b) = \begin{vmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{vmatrix} = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2,$$

and we have the following

1. If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
2. If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
3. If $D(a, b) < 0$, then $f(a, b)$ is a saddle point.

Example 3.7.6 Find the local maximum and local minimum values of the function

$$f(x, y) = x^4 + y^4 - 4xy + 1.$$

Example 3.7.7 Find the shortest distance from the point $(1, 0, -2)$ to the plane $x + 2y + z = 4$.

Definition 3.7.8 If f is a function of two variables, we say that f has a global (or absolute) minimum at (a, b) if $f(x, y) \geq f(a, b)$ for all (x, y) in the domain of f . Similarly, f has a global maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) in the domain of f .

Theorem 3.7.9 (*Extreme Value Theorem*) If f is a continuous function of two variables, defined on a closed and bounded set D , then f attains an absolute maximum and minimum value on D .

To find the maximum and minimum values of a continuous function f on a closed and bounded set D , we do the following

1. Find the values of f at the critical points of f in D .
2. Find the extreme values of f on the boundary of D .
3. Identify the largest and smallest of these values.

Example 3.7.10 Find the absolute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y) | 0 \leq x \leq 3, 0 \leq y \leq 2\}$.